

INTRODUCTION TO C

C IS A

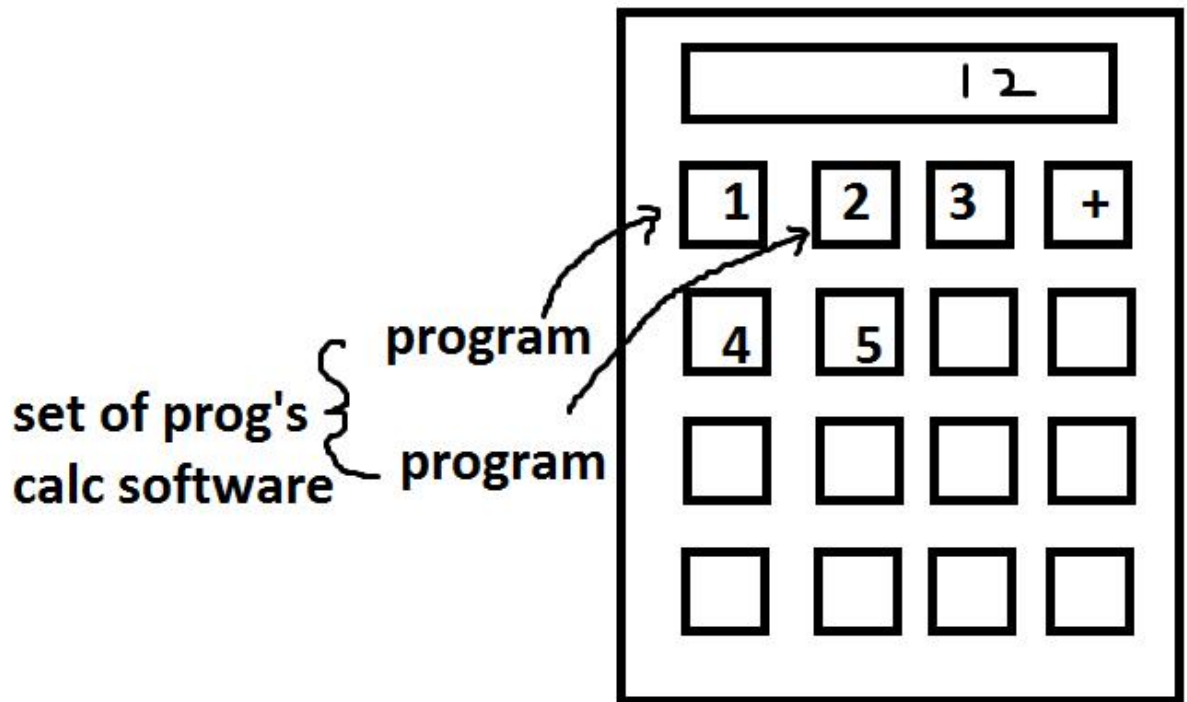
1. It is a high level / middle level programming language.
2. C is a Compiler based programming language.
3. C is a procedure oriented programming [POP] language.
4. C is a general purpose / multipurpose programming language.

What is a program?

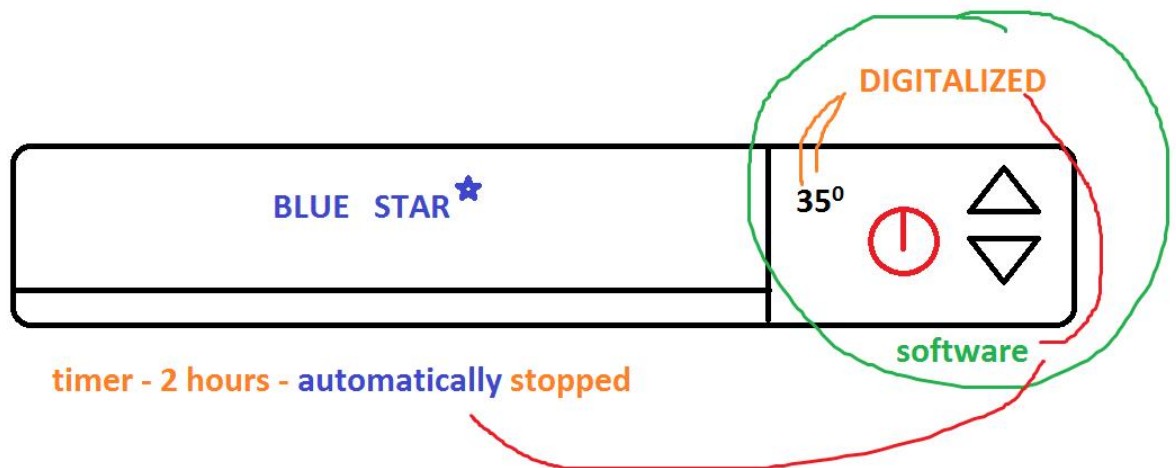
Set of instructions is called program

What is a software?

Set of programs is called software.



As per IT Industry software is a digitalized and automated process.



Basically the software divided into 2 types.

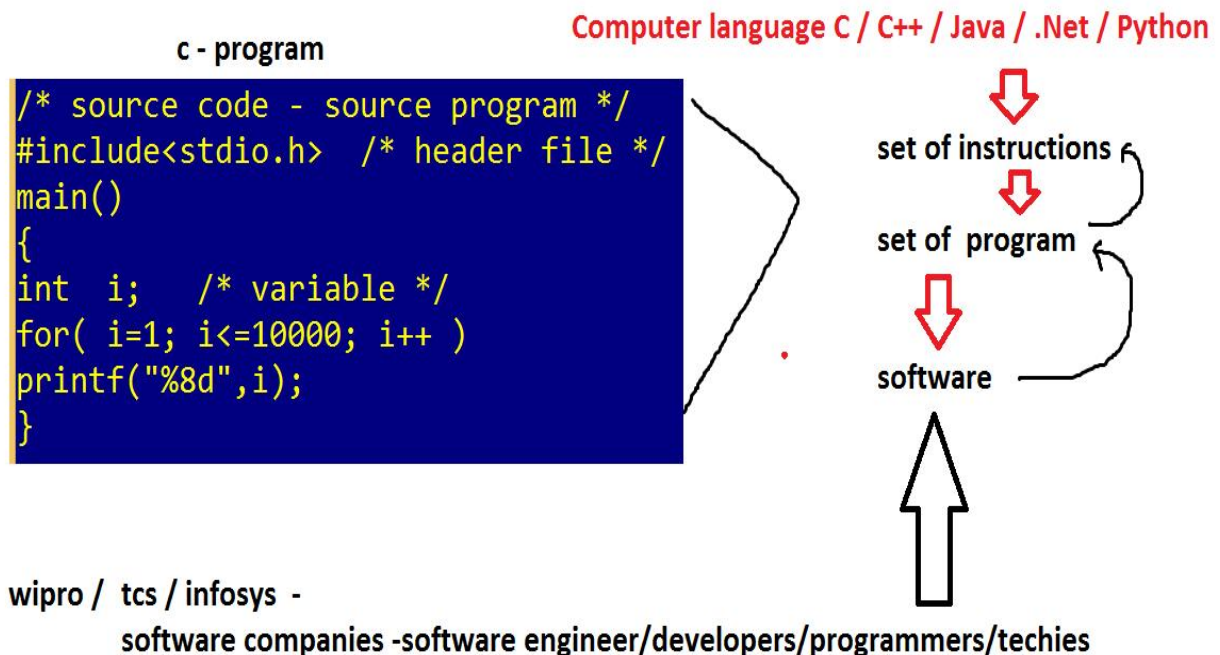
1. System software

Eg: operating system, device drivers, translators

2. Application software

Eg: whatsapp, phonepe, irctc,...

What is a language? Generally the languages like telugu / English / hindi / Marathi etc are human languages and they are used to communicate with the humans. To communicate with the computers we have to use the **computer programming languages** like C / C++ / Java / .Net / Python etc. By using all these languages we are creating the software [programs] to **communicate with the machines**.



Basically the computer languages are divided into 3 types.

1. **Machine language**: Created with binary code[0, 1]

Eg: 10111100

2. **Low level / assembly language**: Created with English like shortcuts called **MNEMONICS**.

Eg: gd mrg, sub, add, ..

The image shows a screenshot of the Turbo C++ IDE. The top window, titled 'ASS.CPP', contains the following C code:

```
[■] ASS.CPP 1
#include<stdio.h>
void main() {
    int a = 10, b = 20, c;

    asm {
        mov ax,a
        mov bx,b
        add ax,bx
        mov c,ax
    }

    printf("c=%d_",c);
}
```

The bottom window shows the output of the program: `c=30_`. The Windows taskbar at the bottom includes icons for various applications and a system tray showing the time as 3:03 PM on 06-Aug-24.

3. High level language: Created with simple english and easy to understand.

Eg: good morning, subject, subtract, subway,
subordinate, subscribe, substitute, addition, address,
addiction, ...

Turbo C++ IDE

File Edit Search Run Compile Debug Project Options Window Help

NONAME00.CPP 2

```
[■]
#include<stdio.h>
void main()
{
int a=10, b=20, c=a+b;
printf("Sum=%d",c);
}
```

Turbo C++ IDE

Sum=30_

C comes with both low level and high level features. Hence it is also called middle level language.

C **low level features** used to develop **system software**.

High level features used to develop **application software**.

Hence c is a **multi-purpose** programming language.

What is a translator?

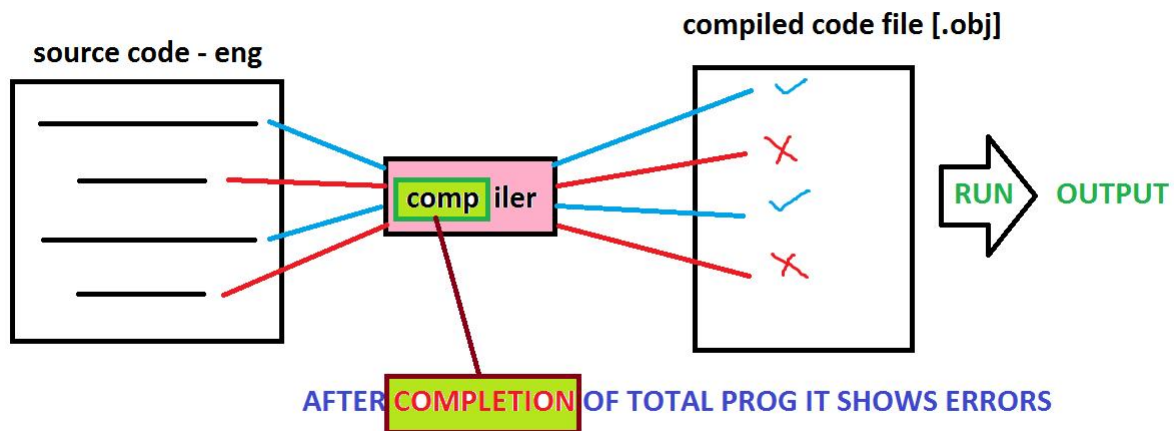


Always the user given instructions are in English, which is called source code / source program. But the computer understandable code is binary code. **To convert this source code into binary code and to check the errors [programming mistakes] we are using the translators like**

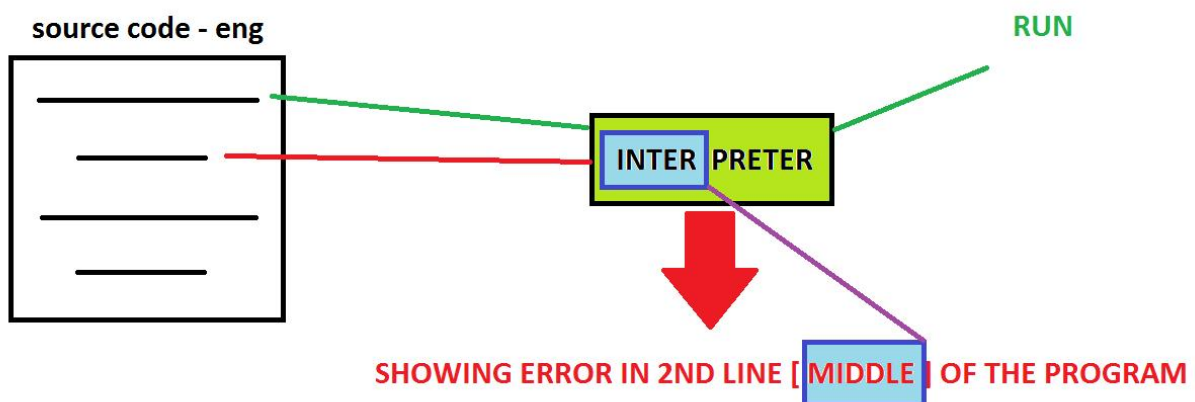
1. Compiler
2. Interpreter
3. Assembler

Compiler and interpreter both used to convert **high level** programs to **machine language**.

Compiler converts the total source code into binary code **at once** by leaving error lines.



Interpreter converts **line by line**.



Assembler used to convert **low level** programs to machine language and working style is similar to compiler.

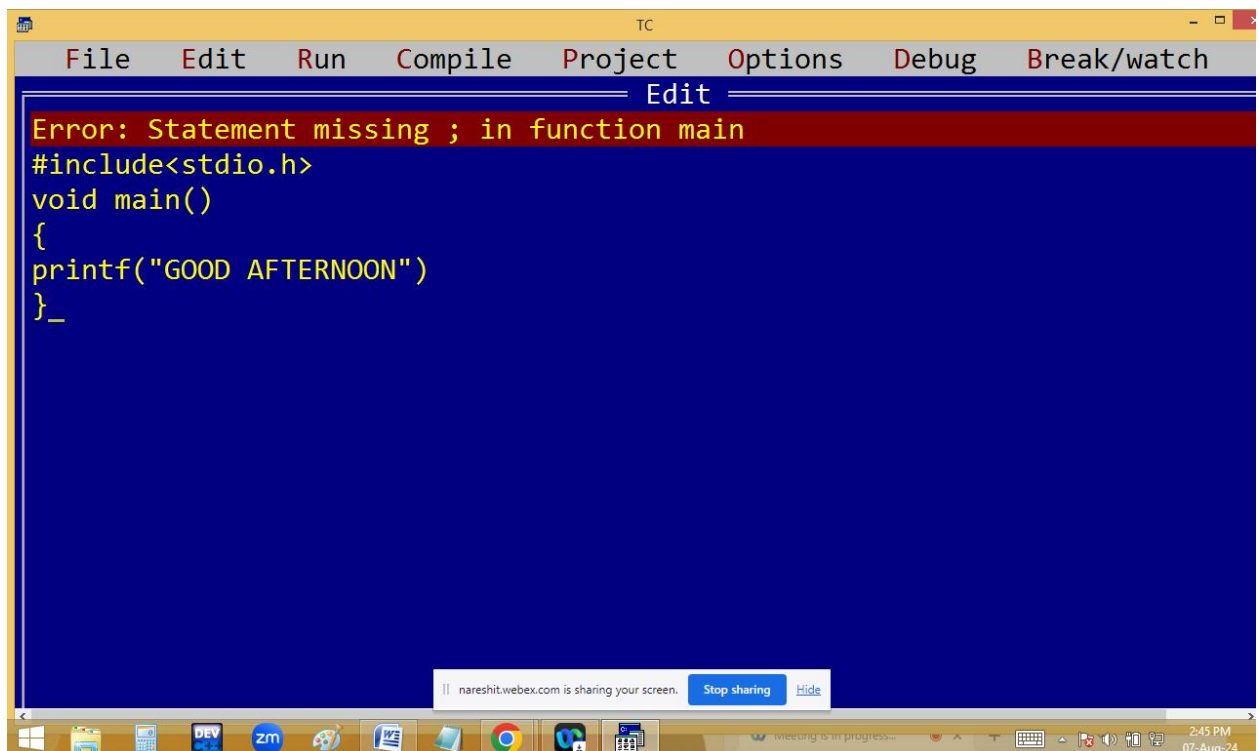
Every **compiler** does two things.

1. Checking errors
2. Source code [eng prg] to binary code

We are having 2 types of errors.

1. Compile time errors / syntax errors

When we are not following programming language rules then it return compile time errors.



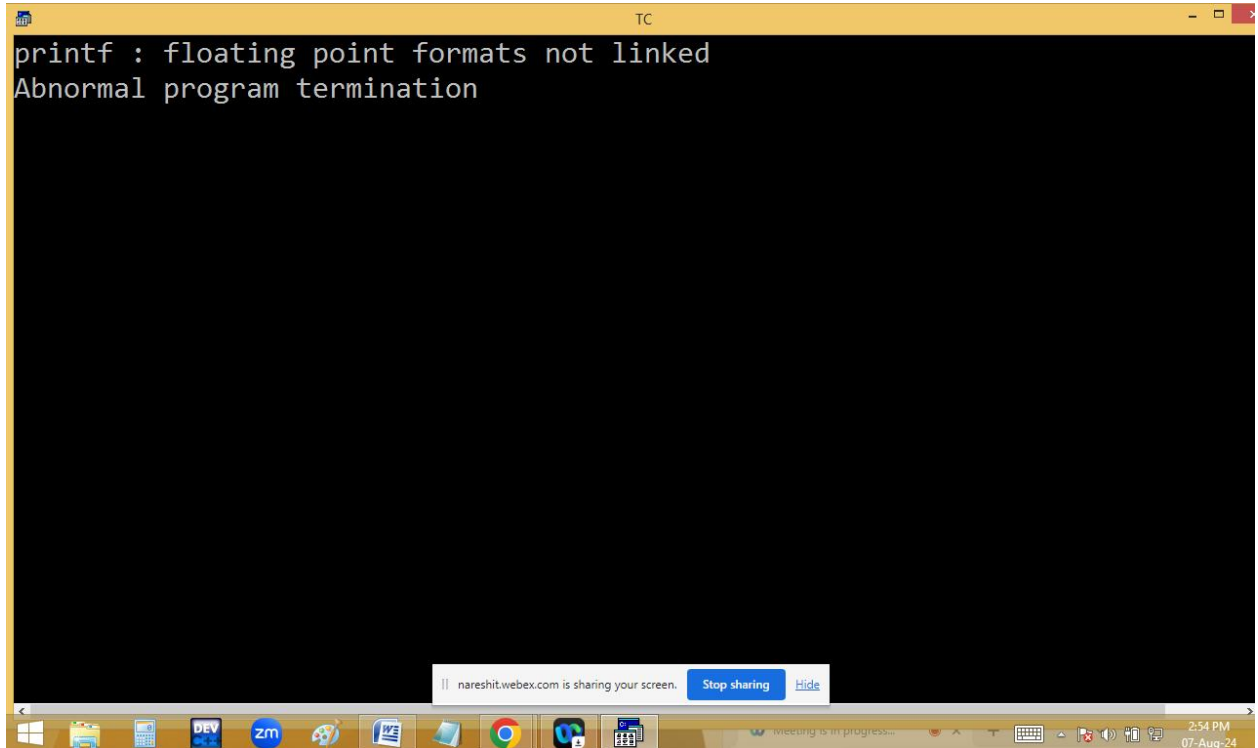
The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The 'Edit' menu is open, and a red error message is displayed: 'Error: Statement missing ; in function main'. The code in the editor is as follows:

```
#include<stdio.h>
void main()
{
printf("GOOD AFTERNOON")
}_
```

The code is missing a semicolon at the end of the printf statement. A watermark at the bottom of the window reads 'nareshit.webex.com is sharing your screen.' with 'Stop sharing' and 'Hide' buttons. The Windows taskbar at the bottom shows various icons and the system clock indicating 2:45 PM on 07-Aug-24.

2. Runtime errors

They are displayed at runtime.



```
printf : floating point formats not linked
Abnormal program termination
```

Warning: system given precautions.

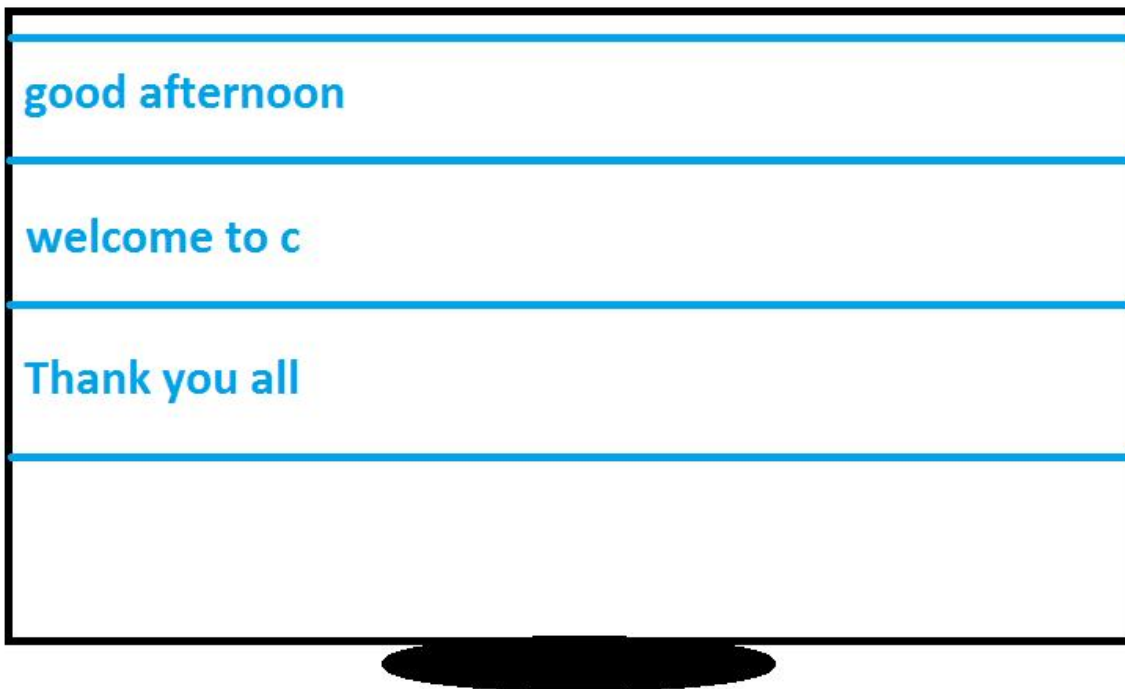
What is a programming paradigm?

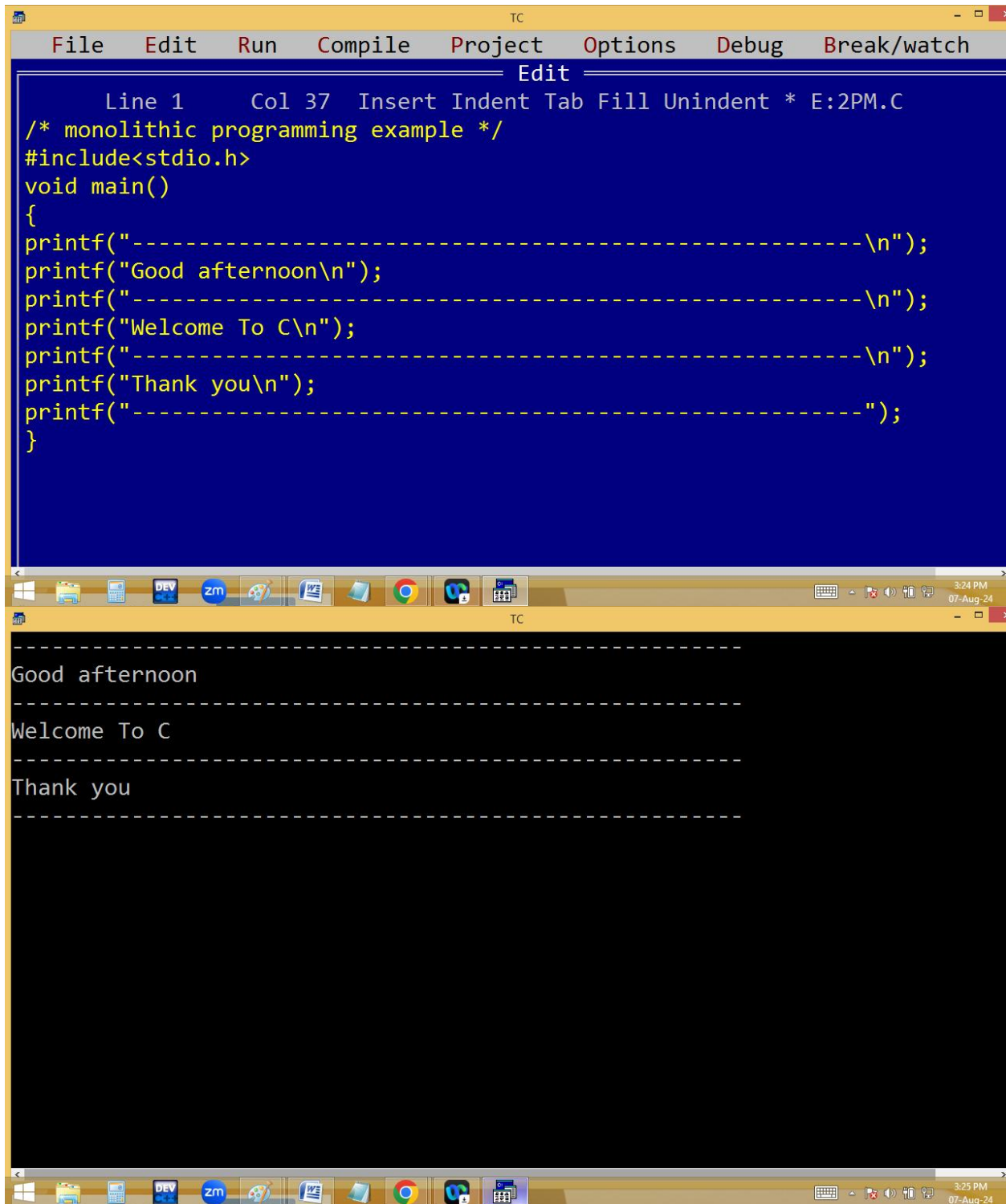
Every programming language comes with certain rules and regulations with a structure, which is called programming paradigm. As a programmer we should have to follow this structure.

Before c language the languages are using **monolithic programming paradigm**. In this the total application developed with single program. Due to this the problems are

1. It is very difficult to find the errors.
2. Takes more memory.
3. Performance is low
4. We can't use part of the program.

Example for monolithic programming:





The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the 'Edit' window, displaying a C program. The program includes a comment, a header file, and a main function that prints several lines of text. The bottom window is the 'Output' window, showing the execution results of the program. The Windows taskbar at the bottom indicates the date as 07-Aug-24 and the time as 3:24 PM and 3:25 PM.

```
Line 1      Col 37  Insert Indent Tab Fill Unindent * E:2PM.C
/* monolithic programming example */
#include<stdio.h>
void main()
{
printf("-----\n");
printf("Good afternoon\n");
printf("-----\n");
printf("Welcome To C\n");
printf("-----\n");
printf("Thank you\n");
printf("-----");
}
```

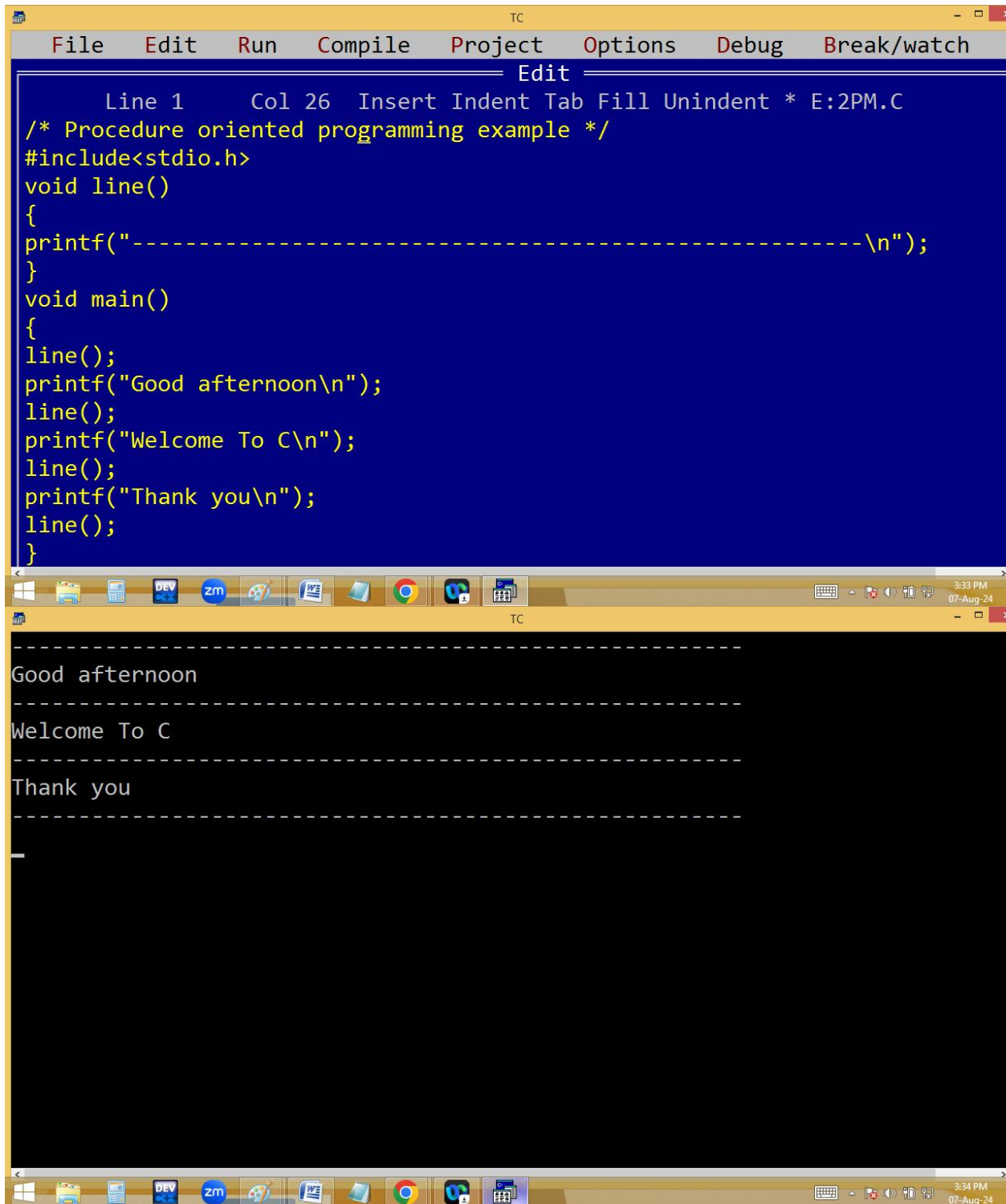
Good afternoon

Welcome To C

Thank you

To avoid all these problems in c language they have introduced procedure oriented programming structure.

In this a big program divided into several small sub programs called procedures / functions / modules / structures. i.e. c program is collection of procedures.



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window, titled 'Edit', displays a C program in a blue editor. The code defines a function 'line()' that prints a dashed line, and a 'main()' function that calls 'line()' and prints 'Good afternoon', 'Welcome To C', and 'Thank you'. The bottom window, titled 'TC', shows the program's output on a black background, displaying the dashed lines and the printed text. The Windows taskbar at the bottom includes icons for various applications and the system clock showing 3:33 PM on 07-Aug-24.

```
Line 1      Col 26  Insert Indent Tab Fill Unindent * E:2PM.C
/* Procedure oriented programming example */
#include<stdio.h>
void line()
{
printf("-----\n");
}
void main()
{
line();
printf("Good afternoon\n");
line();
printf("Welcome To C\n");
line();
printf("Thank you\n");
line();
}
```

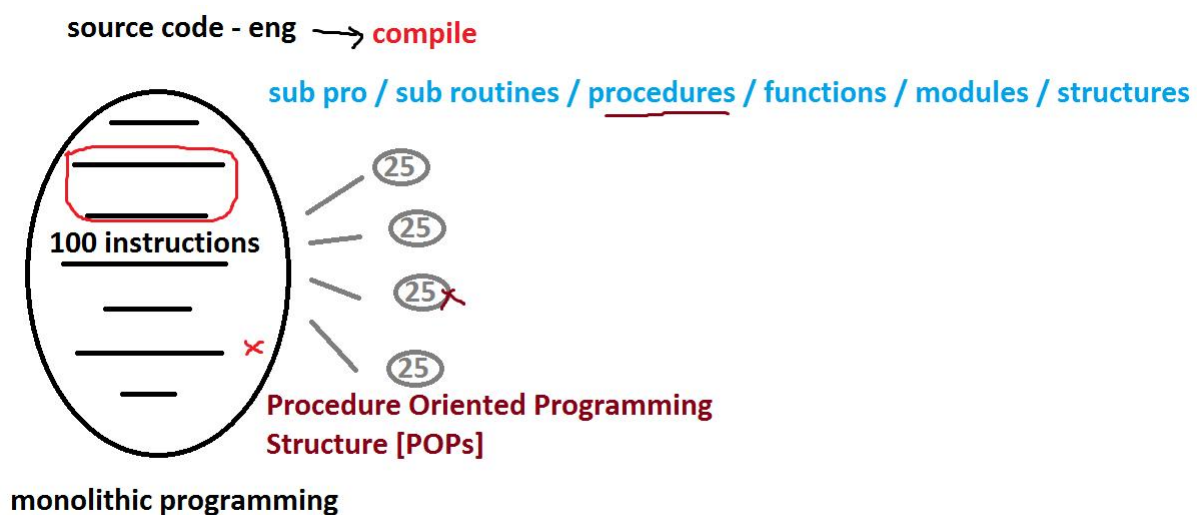
Good afternoon

Welcome To C

Thank you

Advantages of POP:

1. **Modularity:** Dividing big program into small modules [pieces] as per the project requirement.
2. **Simplicity:** easy to read and understand.
3. **Reusability:** write once, use many times.
4. **Efficiency:** performance is high

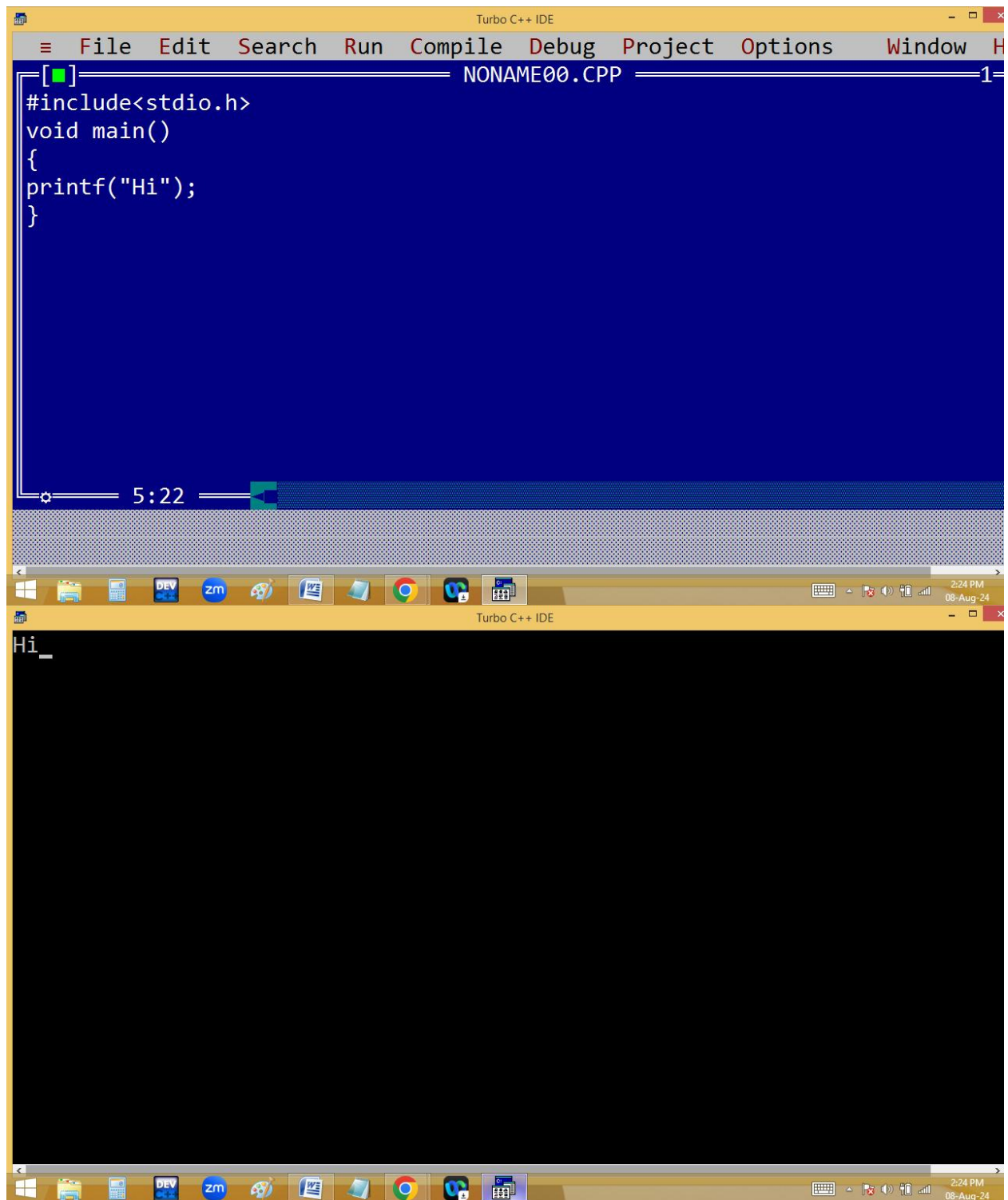


C uses POP's

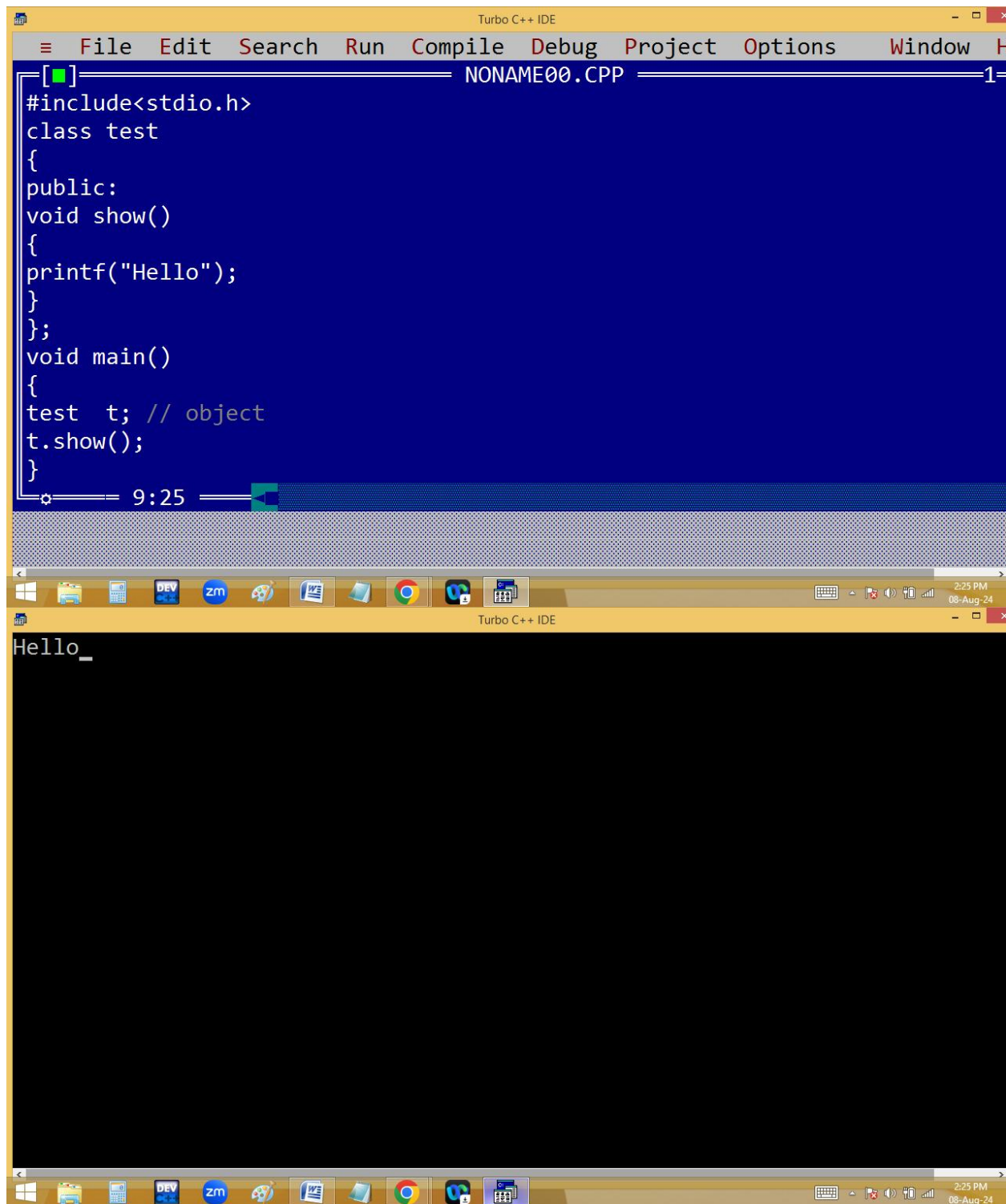
C++ & Python – POP's & OOP's – **Multi paradigm programming languages.**

Java/.net - oops

Pops example:



Oops example:



The image shows a screenshot of the Turbo C++ IDE. The top window, titled "NONAME00.CPP", contains the following C++ code:

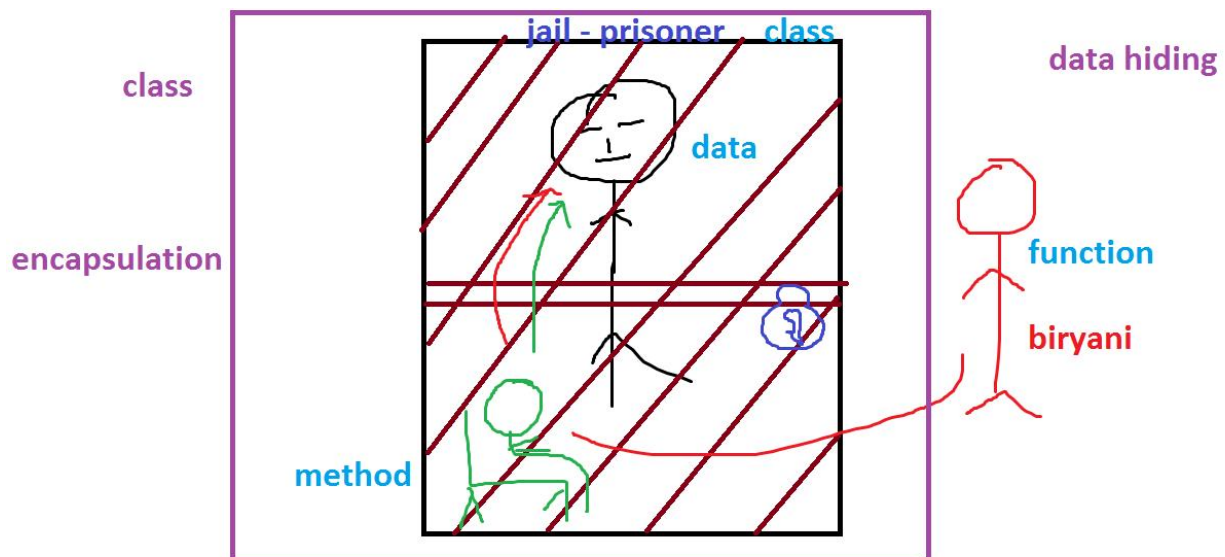
```
[ ] NONAME00.CPP 1
#include<stdio.h>
class test
{
public:
void show()
{
printf("Hello");
}
};
void main()
{
test t; // object
t.show();
}
```

The bottom window shows the output of the program, which is "Hello_". The Windows taskbar at the bottom indicates the time is 2:25 PM on 08-Aug-24.

The disadvantage of pops is data is public.

OOP's features:

1. Class
2. Object
- 3. Data hiding**
- 4. Encapsulation**
- 5. Inheritance**
 - a. Reusability**
 - b. extensibility**
6. Polymorphism ➔ poly means many + morph – forms
/ shapes / kinds
7. Abstraction - briefing



Why c is a general / multipurpose programming language?

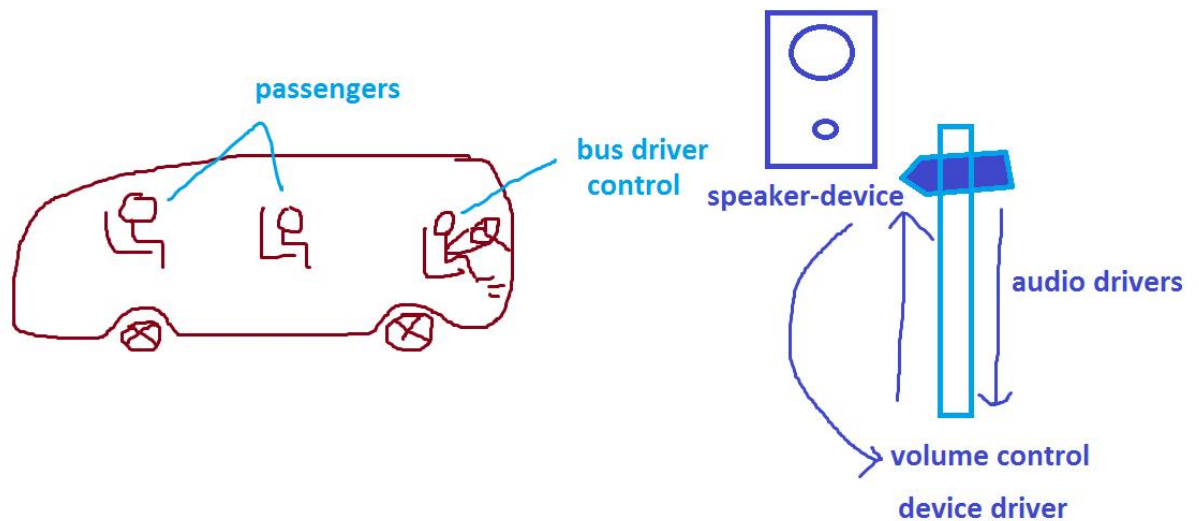
Using C language we can develop the software like

1. Operating system

Eg: windows, unix, linux, android, mac, ios,...

2. Device drivers

Eg : audio / video / usb drivers,.

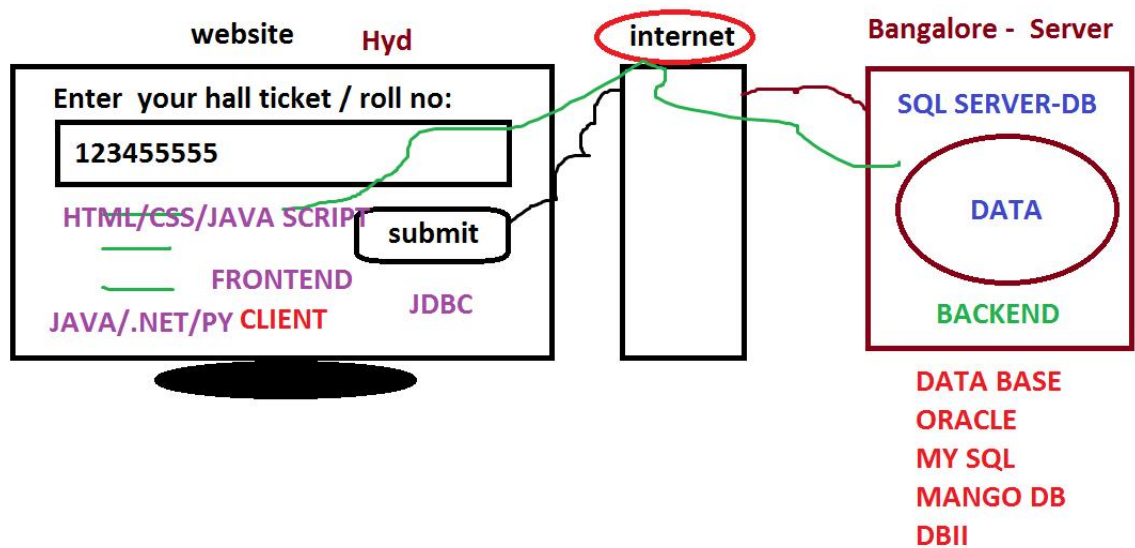


3. Translators

Eg: compiler, interpreter, assembler

4. Data base

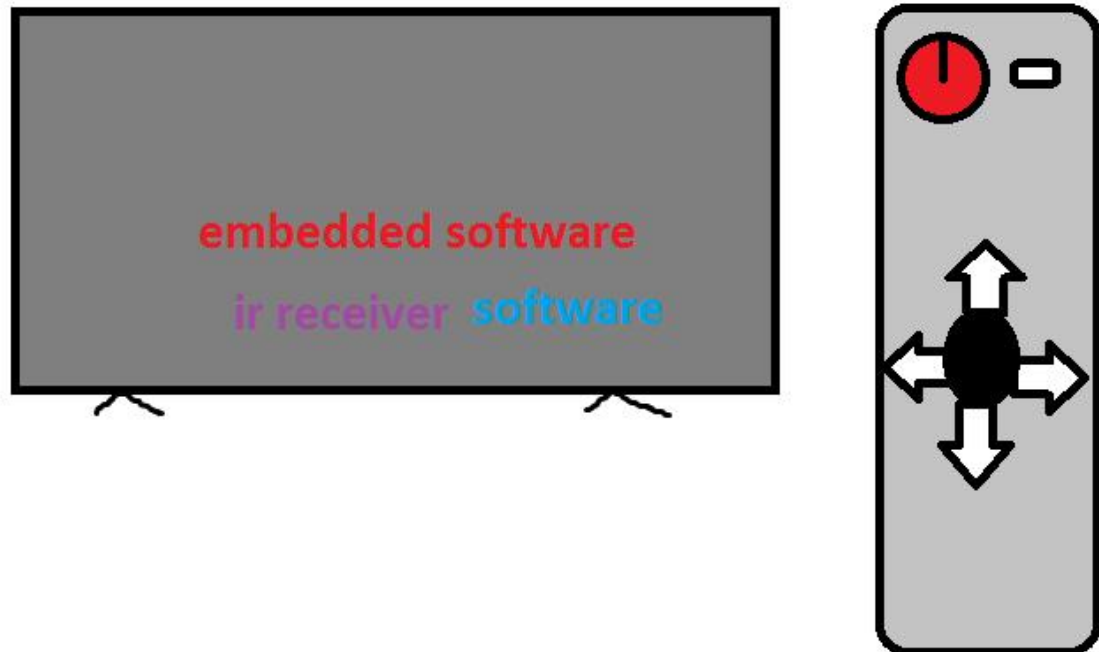
Eg: oracle, mysql, sql server, mongo db, dbII,..

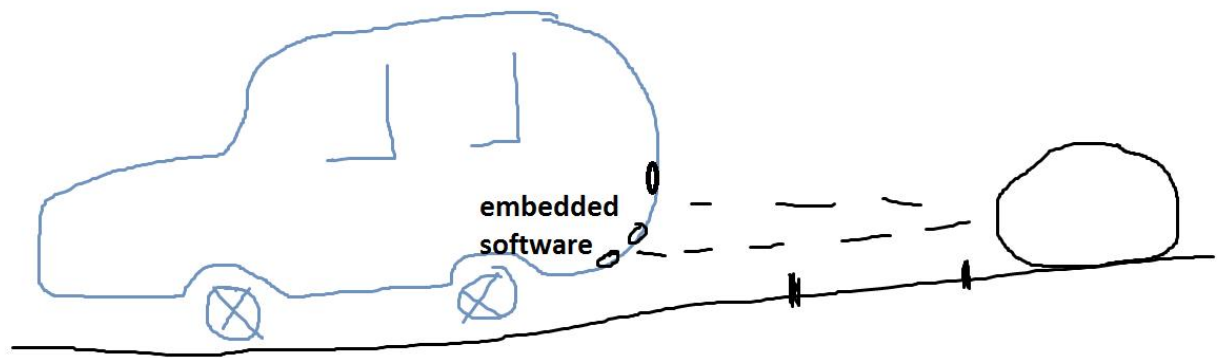


5. Commercial applications:

Eg: hotel, super market, college, phone pe,....

6. Embedded applications





7. Editors

Eg: notepad, wordpad, ms-word,...

8. PC & Mobile games

9. Anti virus

Eg: avast, quick heal, mcafee,.....

10. Media players

Eg: vlc, mx-player, windows media player,..

11. Browsers

Eg: chrome, firefox, edge,...

12. Any type of standalone applications

Hence c is a multipurpose programming language.

